Project Design Phase-II
Data Flow Diagram & User Stories

Date	26 February
Team ID	
Project Name	**An Exploration of Electricity Consumption Patterns**
Maximum Marks	4 Marks

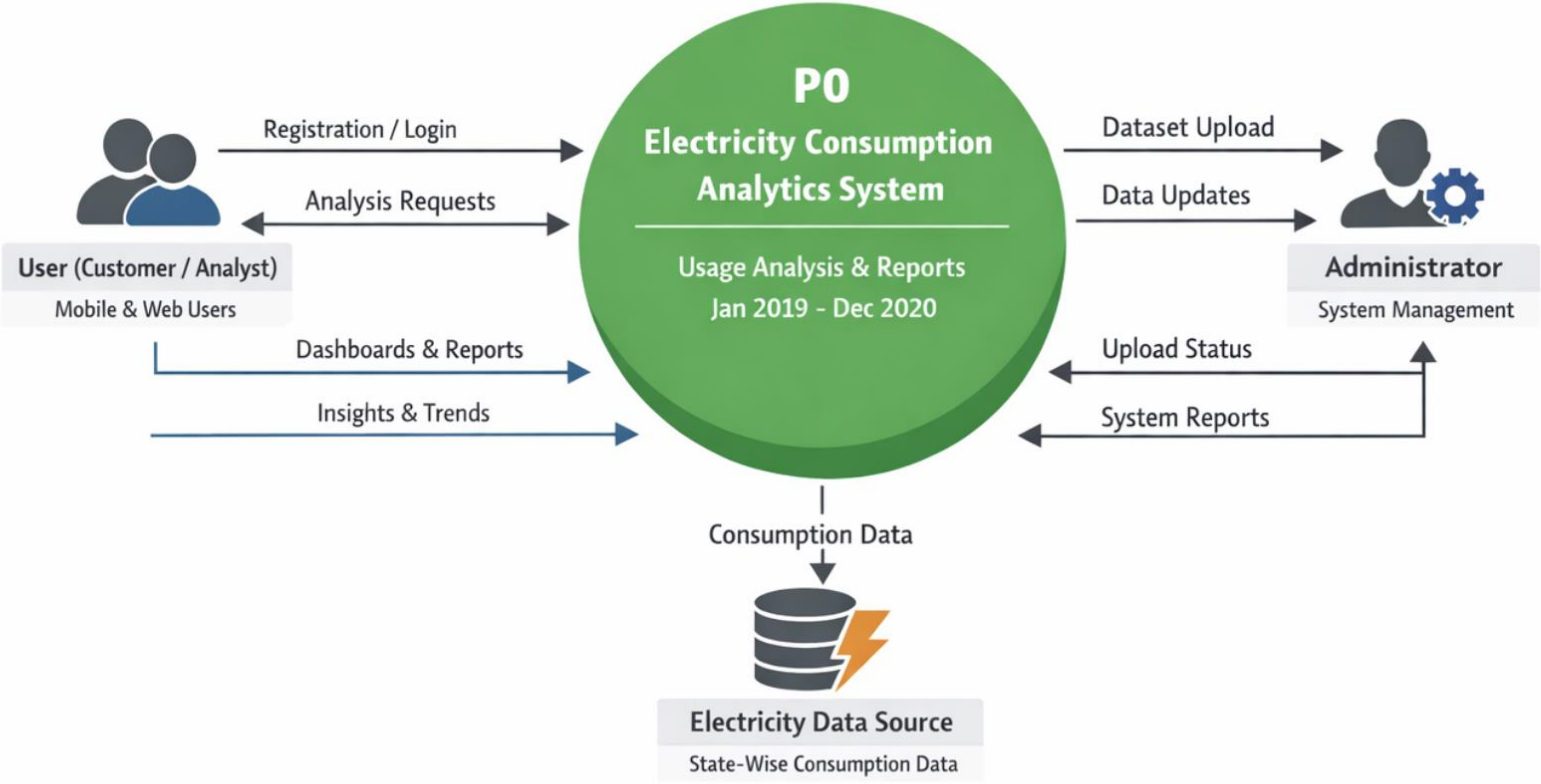
Data Flow Diagrams:

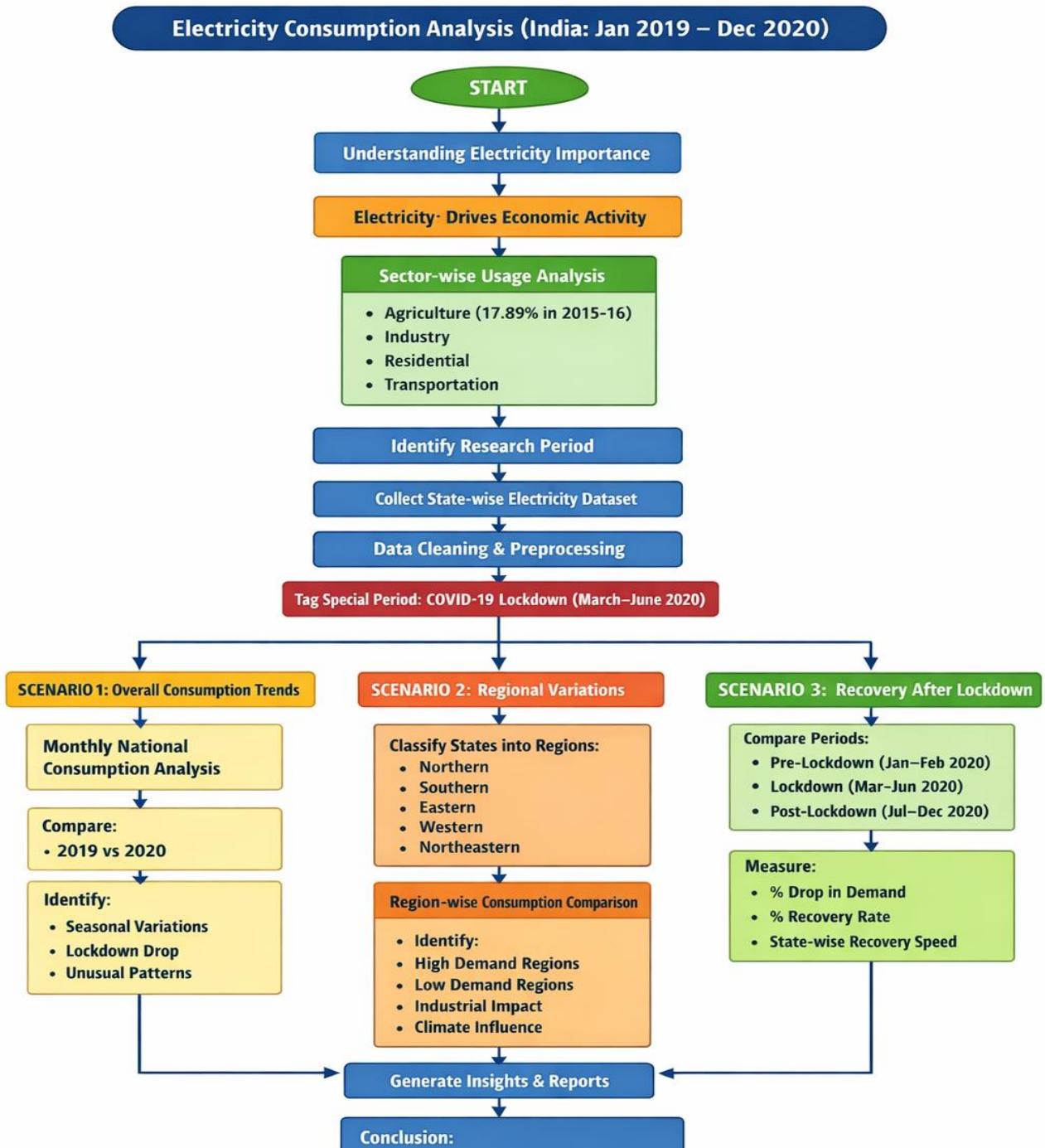
A **Data Flow Diagram (DFD)** is a graphical representation used to show how data moves through a system, explaining the sources of data, the processes that transform it, the data stores where it is kept, and the destinations where it is finally used; it helps in understanding system requirements clearly without focusing on programming details.

DFDs are mainly classified into different levels based on the depth of information they provide: **DFD Level-0 (Context Diagram)** gives a high-level overview of the entire system as a single process interacting with external entities, **DFD Level-1** breaks this main process into major sub-processes and introduces data stores, **DFD Level-2** further decomposes these sub-processes to show detailed data flow and internal operations, and **DFD Level-3**, which is rarely used, provides very fine-grained detail for complex systems.

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level-0: Electricity Consumption Analytics System




Electricity Consumption Analysis (India: Jan 2019 – Dec 2020)
START
Understanding Electricity Importance
Electricity· Drives Economic Activity
Sector-wise Usage Analysis
• Agriculture (17.89% in 2015-16)
• Industry
• Residential
• Transportation
Identify Research Period
Collect State-wise Electricity Dataset
Data Cleaning & Preprocessing
Tag Special Period: COVID-19 Lockdown (March–June 2020)
SCENARIO 1: Overall Consumption Trends
Monthly National Consumption Analysis
Compare:
• 2019 vs 2020
Identify:
• Seasonal Variations
• Lockdown Drop
• Unusual Patterns
SCENARIO 2: Regional Variations
Classify States into Regions:
• Northern
• Southern
• Eastern
• Western
• Northeastern
Region-wise Consumption Comparison
• Identify:
• High Demand Regions
• Low Demand Regions
• Industrial Impact
• Climate Influence
SCENARIO 3: Recovery After Lockdown
Compare Periods:
• Pre-Lockdown (Jan–Feb 2020)
• Lockdown (Mar–Jun 2020)
• Post-Lockdown (Jul–Dec 2020)
Measure:
• % Drop in Demand
• % Recovery Rate
• State-wise Recovery Speed
Generate Insights & Reports
Conclusion:

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Data Analyst	Data Upload	USN-1	As a Data Analyst, I can upload the electricity dataset (2019–2020)	Dataset is successfully uploaded and stored	High	Sprint-1
	Data Cleaning	USN-2	As a Data Analyst, I can clean missing or inconsistent values	Clean dataset available for analysis	High	Sprint-1
	Monthly Trend Analysis	USN-3	As a Data Analyst, I can view month-by-month national consumption	Monthly graph is generated	High	Sprint-1
	Regional Analysis	USN-4	As a Data Analyst, I can compare electricity usage across regions	Region-wise comparison chart available	High	Sprint-2
	Lockdown Impact	USN-5	As a Data Analyst, I can analyze consumption during lockdown period	Lockdown months clearly highlighted in graph	High	Sprint-2
	Recovery Tracking	USN-6	As a Data Analyst, I can compare pre- and post-lockdown consumption	Recovery report generated	Medium	Sprint-3
Government Policy Maker	National Summary	USN-7	As a Policy Maker, I can view overall national electricity trends	Summary dashboard accessible	High	Sprint-3
	Regional Insights	USN-8	As a Policy Maker, I can identify high and low performing regions	Regional comparison report available	High	Sprint-3
	Recovery Status	USN-9	As a Policy Maker, I can see which states recovered faster post-lockdown	Recovery ranking report generated	High	Sprint-3
Public User	Dashboard Access	USN-10	As a Public User, I can view electricity trends through dashboards	Interactive charts available	Medium	Sprint-3
	State Comparison	USN-11	As a Public User, I can compare electricity consumption of two states	Comparison graph generated	Medium	Sprint-3